

# Deepak Mallampati

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## PROFESSIONAL SUMMARY

Multimodal AI engineer with production video-understanding (60-70% review-time reduction via transformer-based summarization) and vision detection (YOLOv8 drowsiness pipeline); LangGraph + Pydantic-structured agentic outputs. Open to ONNX/TensorRT edge inference ramp.

## SKILLS

Multimodal AI / Vision • Foundation Models • LangGraph + AutoGen • Pydantic AI / Structured Outputs • Embeddings + Vector Search • Edge Inference (ramp) • Python • Applied AI / LLMs • Eval Harnesses • Docker / CI-CD

**Languages:** Python, TypeScript, JavaScript, SQL (T-SQL, PostgreSQL, SQLite), Java, C++

**AI / ML:** PyTorch, scikit-learn, XGBoost, LangChain, LlamaIndex, OpenAI, Anthropic, Hugging Face Transformers, Diffusers, YOLOv8

**LLM / GenAI:** RAG, agentic workflows, structured outputs, prompt engineering, evaluation pipelines, guardrails, grounding, embeddings, vector search

**Backend:** FastAPI, Flask, REST, WebSockets, Docker, Celery, Playwright

**Frontend:** React, Next.js, Vite, Tailwind, Chrome Extensions (Manifest V3)

**Data:** PostgreSQL + pgvector, MongoDB, Redis, Pandas, NumPy, Snowflake

**DevOps:** Docker, Jenkins, GitHub Actions, Grafana, Prometheus, AWS (EC2, S3, Lambda)

**Domains:** Healthcare AI (HIPAA-conscious), Applied AI, Enterprise AI, Workflow Automation, Computer Vision, Multimodal

## PROFESSIONAL EXPERIENCE

### Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer Trainee - Agent Engineering, Production LLM

USA - Healthcare Technology, AI Consulting

- Built **Retrieval-Augmented Generation (RAG)** for clinical knowledge retrieval; recursive semantic chunking and transformer embeddings improved retrieval precision ~**35%**, reduced irrelevant retrieval >**30%**, response alignment >**90%**.
- Developed **agentic LLM workflows** for multi-step healthcare queries with tool discipline, structured reasoning, and grounding rules; response stability ~**25%**, hallucinations cut >**30%**.
- Shipped **predictive ML pipelines** (scikit-learn, XGBoost) for patient no-show and care engagement scoring; recall **+15-20%** on high-risk categories, precision held >**90%** via class weighting, stratified sampling, threshold calibration.
- Packaged ML/LLM inference as **FastAPI / Flask** services on **Docker** with structured logging and load simulation; post-deploy defects ~**30%**.
- EHR preprocessing (Pandas, NumPy); dataset reliability >**98%**, downstream instability **-40%**.
- HIPAA-conscious governance: de-identification, data lineage, evaluation audit trails, stakeholder-facing system-limitation docs.

### Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern)

Hyderabad, India - Oil & Gas ERP

- Optimized **T-SQL stored procedures** on a compliance-sensitive ERP (contracts, nominations, allocations, invoicing); query time **-20%**, retrieval latency **-25%**.
- Built **Jenkins CI/CD pipelines** for schema updates and stored-procedure deployments; deploy errors **-30%**, release cycle **-35 to -40%**.
- Designed performance dashboards with **SQL Server DMVs + Grafana**; incident recurrence **-25%**, root-cause resolution **-30%**.
- Tuned index maintenance and partition-aware queries; reconciliation runtime **-15%**, deadlocks **-15%**.
- Supported RBAC and audit logging for financial modules in a regulated environment.

*Machine Learning Engineer**Hyderabad, India - Nonprofit, Video Analytics, Applied NLP*

- Built **transformer-based video summarization** over 5,000+ recorded sessions; hierarchical summarization and timestamp-aligned clip extraction cut manual review **60-70%**.
- AI-selected highlights aligned with human-curated moments **~85%**.
- Implemented **document Q&A with RAG**; hallucinations **~30%**, contextual accuracy **>85%**.
- Deployed as **Flask** API with a lightweight web interface for non-technical staff.

## PROJECTS

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### **YOLOv8 Driver Drowsiness Detection**

Real-time computer vision for driver drowsiness; sub-50ms inference on edge hardware. Walk-forward validation and class-weighted training for imbalanced safety data.

*PyTorch, YOLOv8, OpenCV, ONNX*

### **Dream Decoder - Multimodal Creative Platform**

End-to-end multimodal pipeline: text description, intermediate prompt-transformation layer, grounded image generation. ~30% contextual alignment gain, ~25-30% first-pass image success.

*FastAPI, React + Vite, Tailwind, Perplexity Sonar, Replicate*

### **Agentic Healthcare Claims Processing & Fraud Risk Intelligence System**

5-stage multi-agent pipeline (Intake, Validation, Consistency, Duplicate, Risk Scoring) with schema-validated JSON contracts between agents to prevent cascading hallucinations. RAG-grounded CPT/ICD validation against vector-indexed policies.

Audit-ready explainable risk scoring.

*Python, LangChain, FastAPI, pgvector, GPT-4, schema contracts*

### **Manga Lens - Multi-Provider AI Vision Chrome Extension**

Real-time manga translation extension shipped to the Chrome Web Store. Multi-provider abstraction (4 AI vision providers: Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama) with per-provider payload handling for worst-case failure isolation. 29 sites supported; 7-day cache; privacy policy and narrowed permissions.

*TypeScript, Manifest V3, Service Workers, OpenAI / Anthropic / Ollama*

## EDUCATION

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**Master of Science, Computer Science - Kent State University***Aug 2023 - May 2025*

Focus: Applied Machine Learning, NLP, Database Systems.

**Bachelor of Technology, Computer Science - KL University***Jun 2019 - May 2023*

Founder / Lead of E-Farming platform (university project, pilot with 3 cooperatives).

## CERTIFICATIONS

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Deep Learning Specialization - *DeepLearning.AI (Coursera)*

*2024*